

EXPERIMENT – 3

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1.COMMAND LINE INTERFACE

PYTHON CODE:

```
def grade_calculator():  
    print("Student Grade Calculator")  
  
    m1 = float(input("Enter marks for Subject 1: "))  
    m2 = float(input("Enter marks for Subject 2: "))  
    m3 = float(input("Enter marks for Subject 3: "))  
  
    total = m1 + m2 + m3  
    average = total / 3  
  
    print("Total Marks:", total)  
    print("Average:", average)  
  
    if average >= 90:  
        print("Grade: A")  
    elif average >= 75:  
        print("Grade: B")  
    elif average >= 50:  
        print("Grade: C")  
    else:  
        print("Grade: Fail")
```

```
if __name__ == "__main__":  
    grade_calculator()
```

OUTPUT:

```
PS C:\Users\babua\OneDrive\Desktop\240701118> python -u "c:\Users\babua\OneDrive\Desktop\240701118\cl  
i1.py"  
Student Grade Calculator  
Enter marks for Subject 1: 90  
Enter marks for Subject 2: 88  
Enter marks for Subject 3: 58  
Total Marks: 236.0  
Average: 78.66666666666667  
Grade: B
```

2.GRAPHICAL USER INTERFACE

PYTHON CODE:

```
import tkinter as tk  
  
from tkinter import messagebox
```

```
def calculate():  
    try:  
        m1 = float(entry1.get())  
        m2 = float(entry2.get())  
        m3 = float(entry3.get())  
  
        total = m1 + m2 + m3  
        avg = total / 3  
  
        if avg >= 90:  
            grade = "A"  
        elif avg >= 75:  
            grade = "B"  
        elif avg >= 50:  
            grade = "C"  
        else:
```

```
grade = "Fail"
```

```
result_label.config(text=f"Total: {total} Average: {avg:.2f} Grade: {grade}")
```

```
except:
```

```
    messagebox.showerror("Error", "Enter valid marks!")
```

```
root = tk.Tk()
```

```
root.title("Grade Calculator")
```

```
root.geometry("350x250")
```

```
tk.Label(root, text="Subject 1 Marks").pack()
```

```
entry1 = tk.Entry(root)
```

```
entry1.pack()
```

```
tk.Label(root, text="Subject 2 Marks").pack()
```

```
entry2 = tk.Entry(root)
```

```
entry2.pack()
```

```
tk.Label(root, text="Subject 3 Marks").pack()
```

```
entry3 = tk.Entry(root)
```

```
entry3.pack()
```

```
tk.Button(root, text="Calculate", command=calculate).pack(pady=10)
```

```
result_label = tk.Label(root, text="Result: ")
```

```
result_label.pack()
```

```
root.mainloop()
```

OUTPUT:

```
PS C:\Users\babua\OneDrive\Desktop\240701118> python -u "c:\Users\babua\OneDrive\Desktop\240701118\gui1.py"
```

Grade Calculator

Subject 1 Marks
100

Subject 2 Marks
88

Subject 3 Marks
59

Calculate

Total: 243.0 Average: 81.00 Grade: B

3.VOICE USER INTERFACE

PYTHON CODE:

```
import speech_recognition as sr
```

```
def calculate_grade(command):
```

```
    try:
```

```
        words = command.split()
```

```
        m1 = float(words[1])
```

```
        m2 = float(words[2])
```

```
        m3 = float(words[3])
```

```
        total = m1 + m2 + m3
```

```
        avg = total / 3
```

```
if avg >= 90:
    grade = "A"
elif avg >= 75:
    grade = "B"
elif avg >= 50:
    grade = "C"
else:
    grade = "Fail"
```

```
print("Total:", total)
print("Average:", avg)
print("Grade:", grade)
```

```
except:
    print("Say: Marks 80 75 90")
```

```
def listen():
    r = sr.Recognizer()
    with sr.Microphone() as source:
        print("Listening...")
        audio = r.listen(source)
```

```
try:
    command = r.recognize_google(audio).lower()
    print("You said:", command)
    calculate_grade(command)
```

```
except:
```

```
print("Could not understand.")
```

```
if __name__ == "__main__":
```

```
    listen()
```

OUTPUT:

```
PS C:\Users\babua> python -u "c:\Users\babua\OneDrive\Desktop\240701118\vui.py"
Listening...
You said: marks 55 92 75
Total: 222.0
Average: 74.0
Grade: C
PS C:\Users\babua>
```